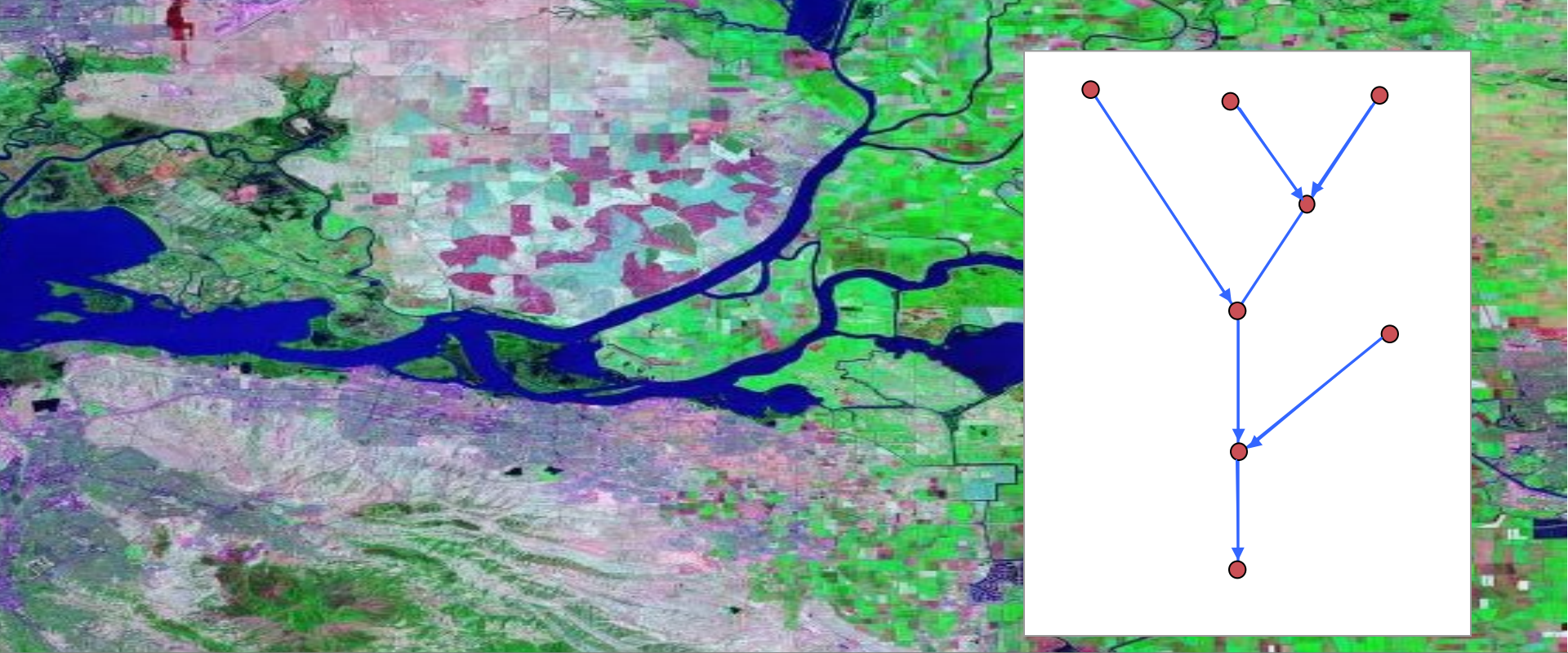




Spatial Data Processing Using SSNbler & GIS

Erin E. Peterson

EP Consulting, Brisbane Australia

Overview

- **Software requirements**
- **Spatial data requirements**
- **Spatial Data Processing Steps**
 - Building the Landscape Network (LSN)
 - Calculating spatial data using the LSN
 - Assembling the SSN object

Software Requirements

- **R Statistical Software version ≥ 3.5**
- **SSNbler version $\geq 1.0.1$**
- **GIS software of your choice**
 - QGIS Version ≥ 3.34 Long Term Release (LTR)
 - ArcPro with the Spatial Analyst extension

Spatial Data Requirements

Data	Format	Description	Required
Streams	Vector	Streams	Yes
Observed sites	Vector	Sites where data were collected	Yes
Prediction sites	Vector	Sites where predictions will be made	No

Spatial data file type

- Imported into R as sf object with LINESTRING or POINT geometry

GeoPackage (.gpkg)

- Open-source, non-proprietary, platform-independent data format for GIS
- SQLite 3 database file with specific database scheme and constraints
- Can store data in multiple formats (e.g. raster, vector) as 'layers'
- Does not have formatting limitations of shapefiles

Spatial Data Requirements

Assumptions about input data

- Observed data stored as points, with one observation per point feature
- Prediction sites stored as points, with one point feature per prediction
- Covariates are stored as attributes in both point datasets

See 'Preparing Input Data for Spatial Stream Network Modeling' tutorial for tips.

	FID	Shape *	OBSPRED_ID	SITE_ID	SampleYear	MeanAugT
1	0	Point	1	303	2001	9.107419
2	1	Point	2	303	2002	12.709355
3	2	Point	3	303	2003	8.77129
4	3	Point	4	303	2004	8.351724
5	4	Point	5	303	2006	8.603793
6	5	Point	6	303	2007	8.955161
7	7	Point	8	303	2012	8.197742
8	8	Point	9	303	2014	8.603548
9	9	Point	10	303	2020	9.592667
10	10	Point	11	9402	1994	11.782903
11	11	Point	12	9402	2001	14.899032
12	12	Point	13	9402	2002	9.290323
13	13	Point	14	9402	2004	9.830667
14	14	Point	15	9402	2014	9.257419

Spatial Data Processing Workflow

Build the Landscape Network



Calculate Spatial Data



Assemble the SSN

Spatial Data Processing Workflow

Build the Landscape Network



Calculate Spatial Data



Assemble the SSN

Landscape Network (LSN)

Hydrologic Modelling Framework

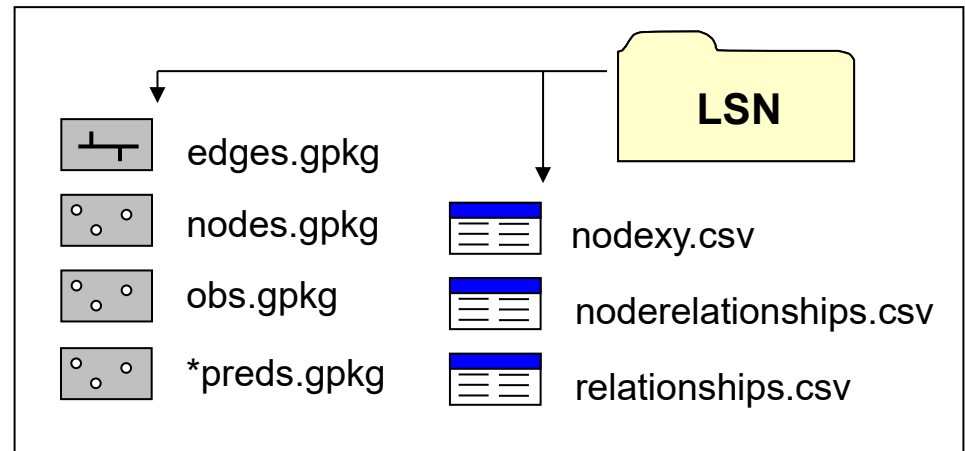
- Data structure utilized by SSNbler
- Spatial data stored as GeoPackages (local files) and sf objects (in memory)
- CSV files store topological relationships of lines and points
 - Measures of connectivity & proximity
 - Retains direction

5 essential elements of a LSN

1. nodes
2. edges
3. nodexy table
4. noderelationship table
5. relationships table

Additional elements

- Observed sites (essential)
- Prediction sites (optional)



*optional elements

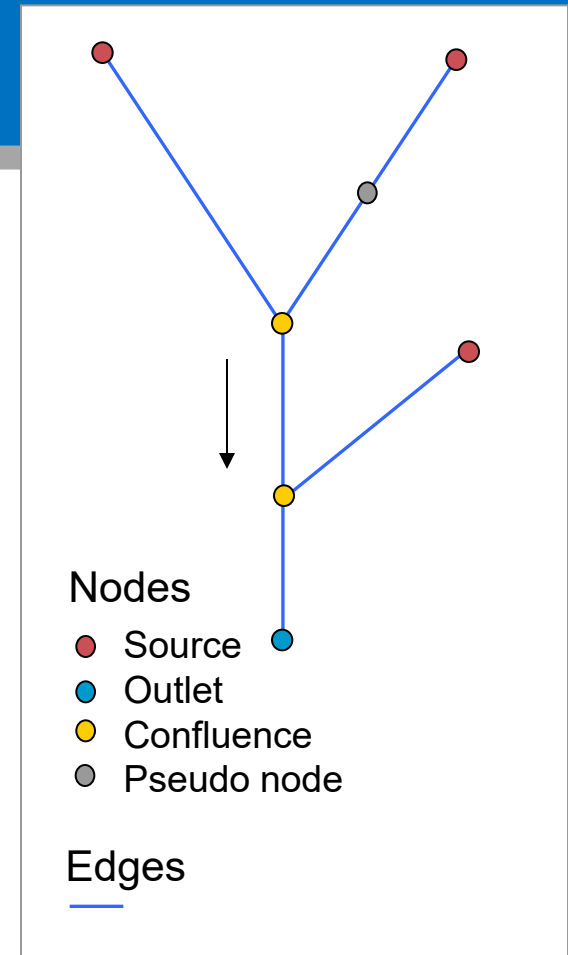
Building the Landscape Network

1. Nodes: topologic breaks in the GIS dataset

- **nodecat**: Sources, outlets, confluences, or pseudo nodes
- Each node is assigned a **pointid**

2. Edges: flow paths from node to node

- **rid** = Reach Identifier: assigned to edges when a LSN is created



pointid	xcoord	ycoord
1	-1528910.943	927495.7884
2	-1528209.763	929590.9821
3	-1529312.422	927012.3763
4	-1530773.659	926078.4378

3. nodexy table

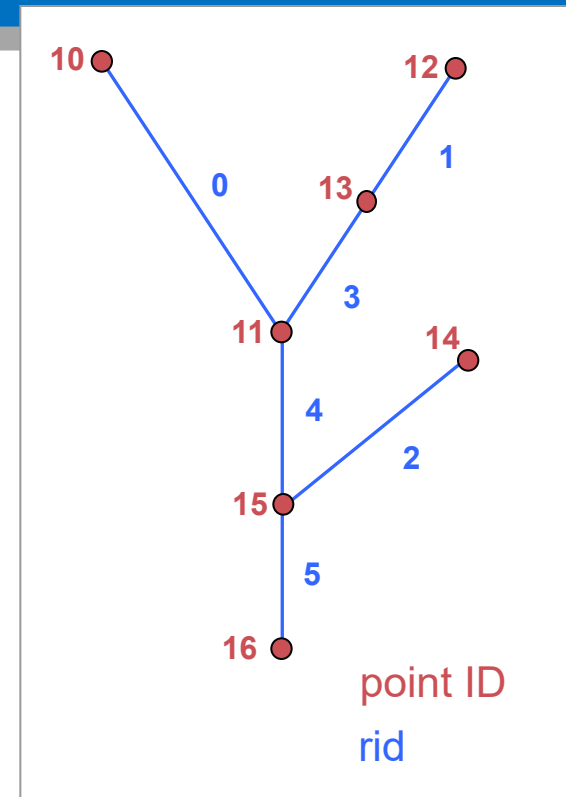
- Provides x,y coordinates for each node

Building the Landscape Network

4. noderelationships table:

- Links nodes to their associated edges
- Based on the edge rid field and node point ID
- Maintains direction

rid	fromnode	tonode
0	10	11
1	12	13
2	14	15
3	13	11
4	11	15
5	15	16

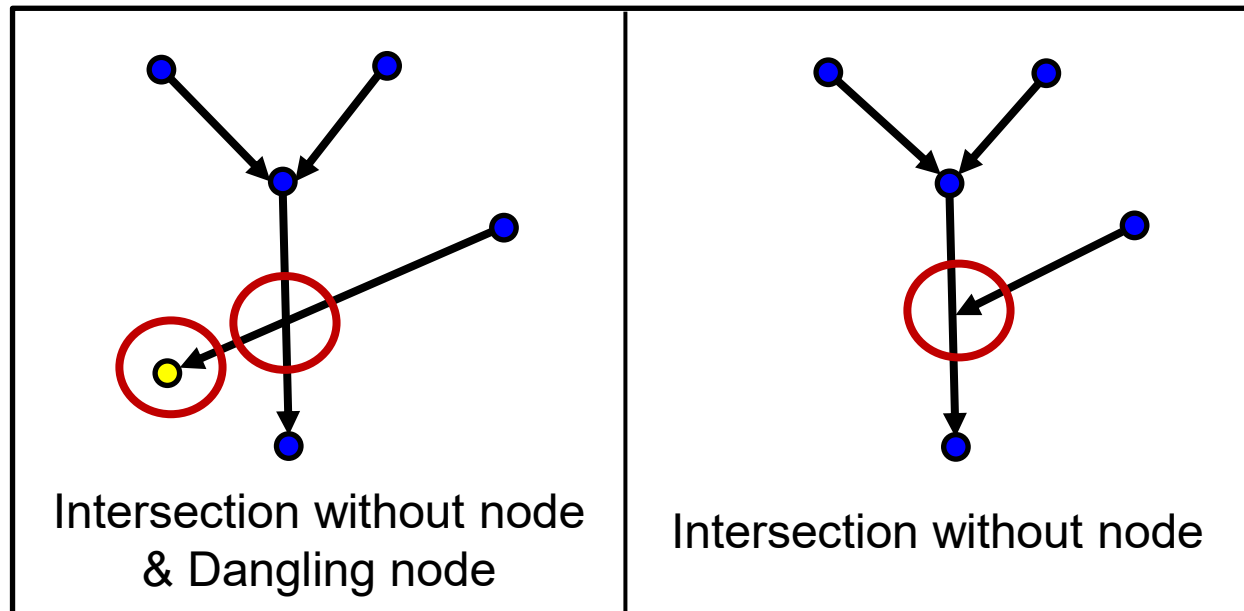
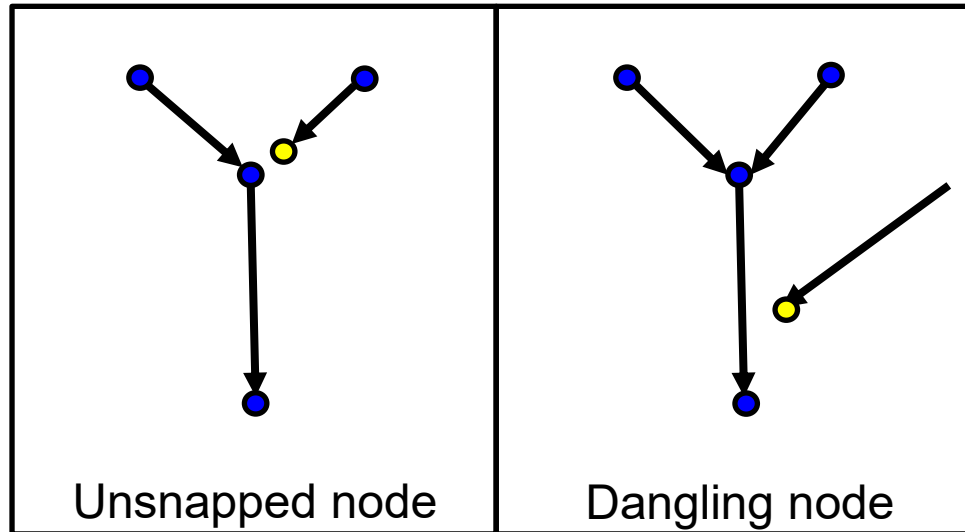


5. relationships table:

- Links edges to one another in a downstream order
- Based on the edge rid

fromedge	toedge
0	4
1	3
2	5
3	4
4	5

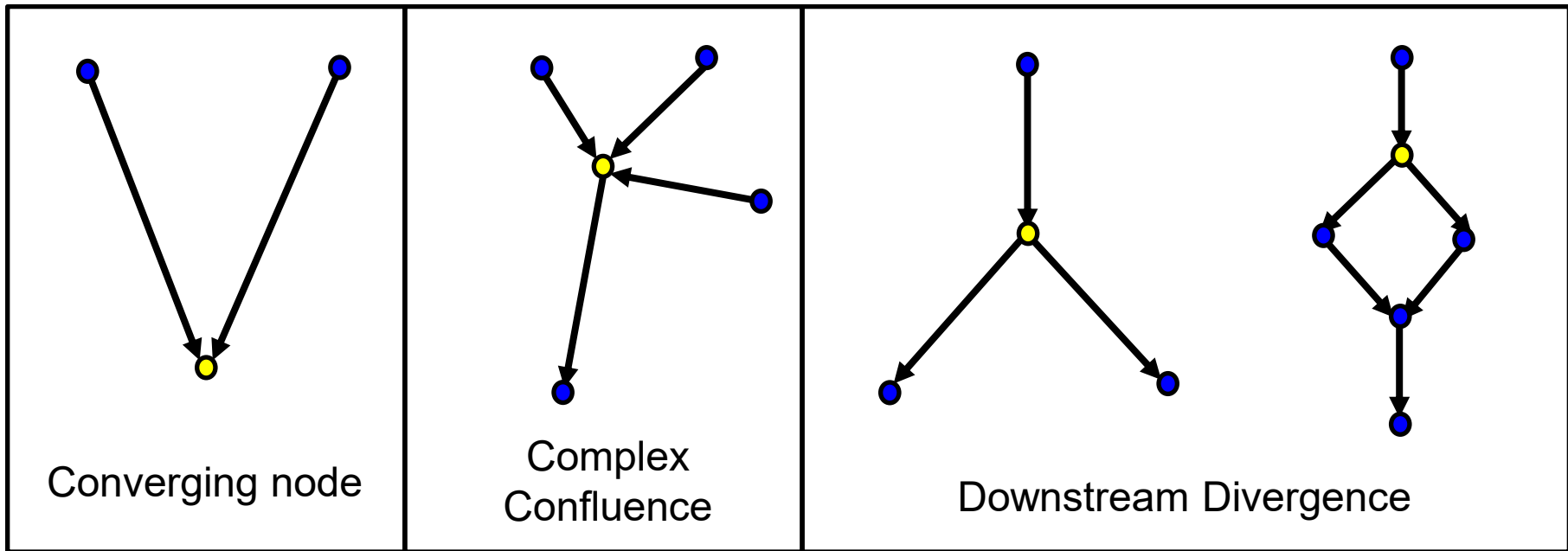
Check & Correct Topology Errors



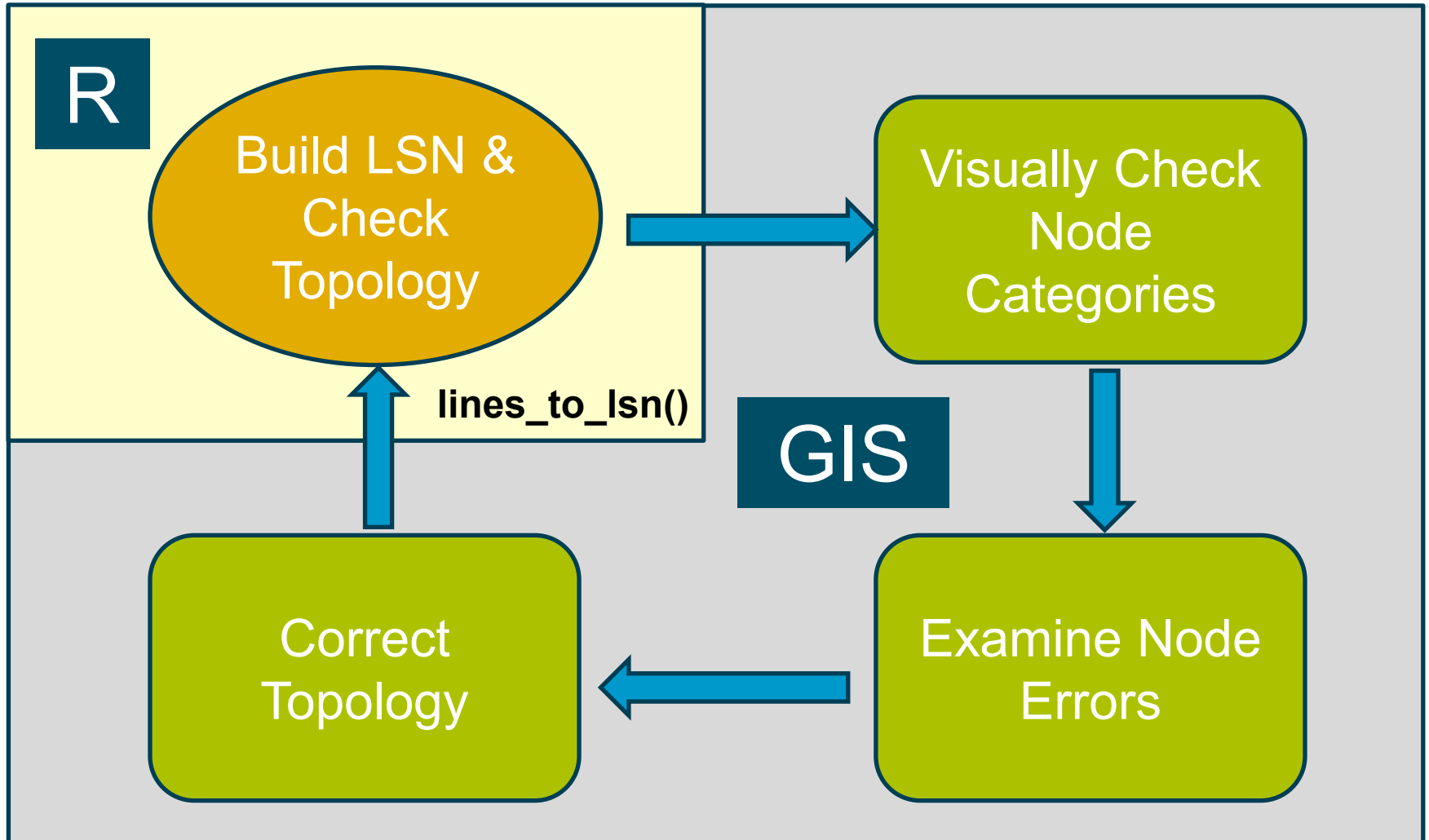
Check & Correct Topological Restrictions

Topological Restrictions

- Three unique relationships not permitted in SSN



Check & Correct Topology



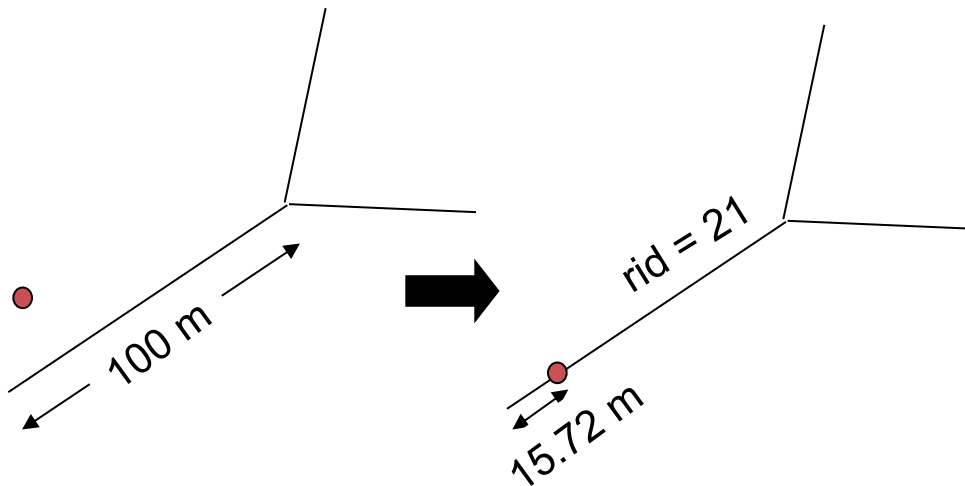
For tips, see topology editing tutorials for QGIS and ArcPro

Building the Landscape Network

Incorporating Sites into the LSN

Sites moved and important columns added: sites_to_lsn()

- Site moved to closest location on an edge feature and **rid** assigned
- Distance moved recorded in **snapdist** column
- Distance **ratio** from end of an edge calculated and assigned



rid	ratio	snapdist
17	0.8122393	0.55209
19	0.5497702	7.40108
21	0.1572009	19.74023

$$15.72 \text{ m} / 100 \text{ m} = 0.1572$$

*** Never skip this step**

Building the Landscape Network

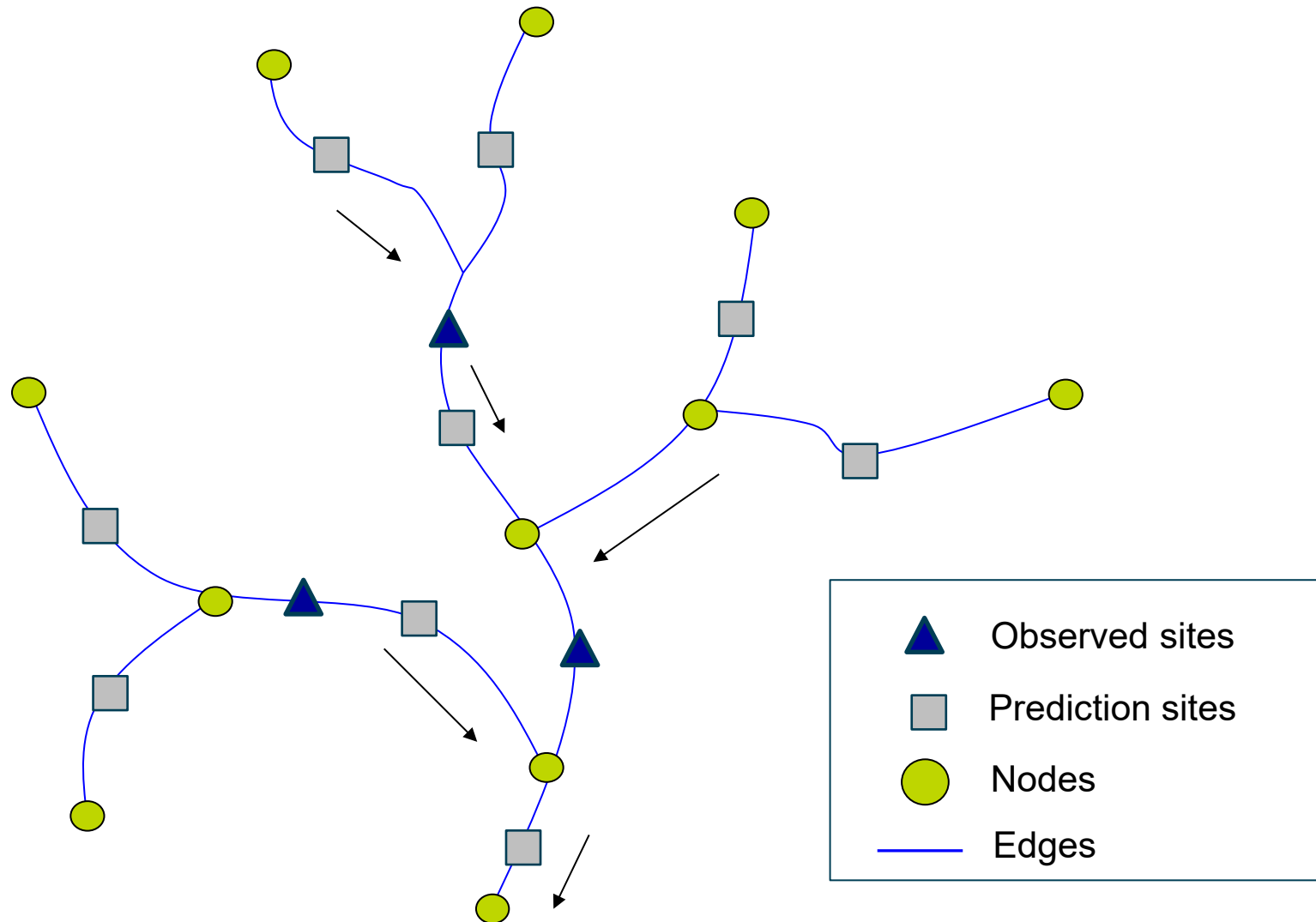
Where is the LSN stored?

LSN Component	In Memory	Local LSN Directory
edges	sf object, LINESTRING geometry	geopackage
observed sites	sf object, POINT geometry	geopackage
prediction sites (optional)	sf object, POINT geometry	geopackage
nodes		geopackage
nodexy table		csv file
noderelationships table		csv file
relationships table		csv file

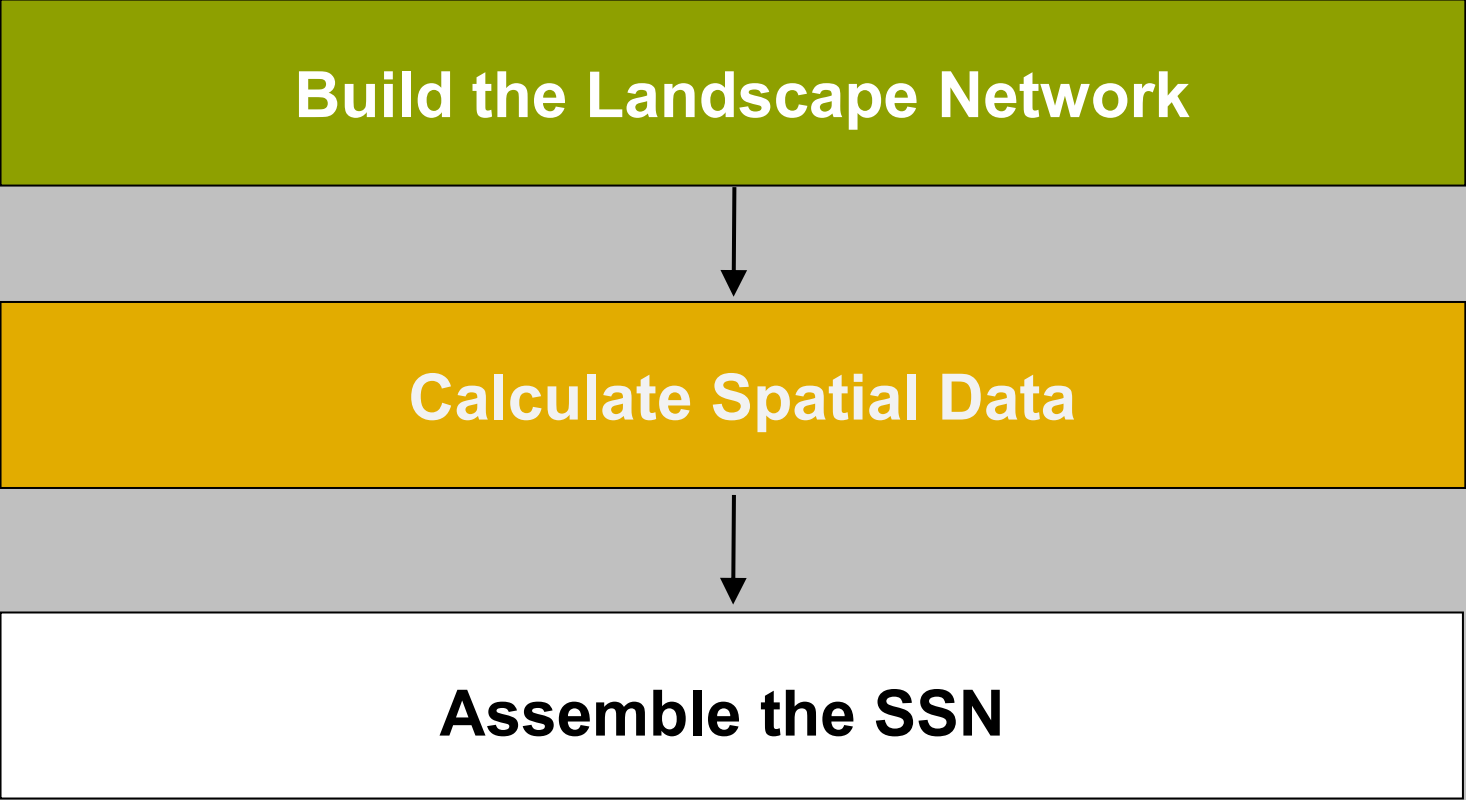
Saving to the local LSN directory

- Initial output files from `lines_to_lsn` are saved to local files (edges.gpkg and csv files)
- Sites Geopackages saved if `save_local = TRUE` (default)
- Updates to columns in GeoPackages saved when `save_local = TRUE` (the default)

Building the Landscape Network



Spatial Data Processing Workflow



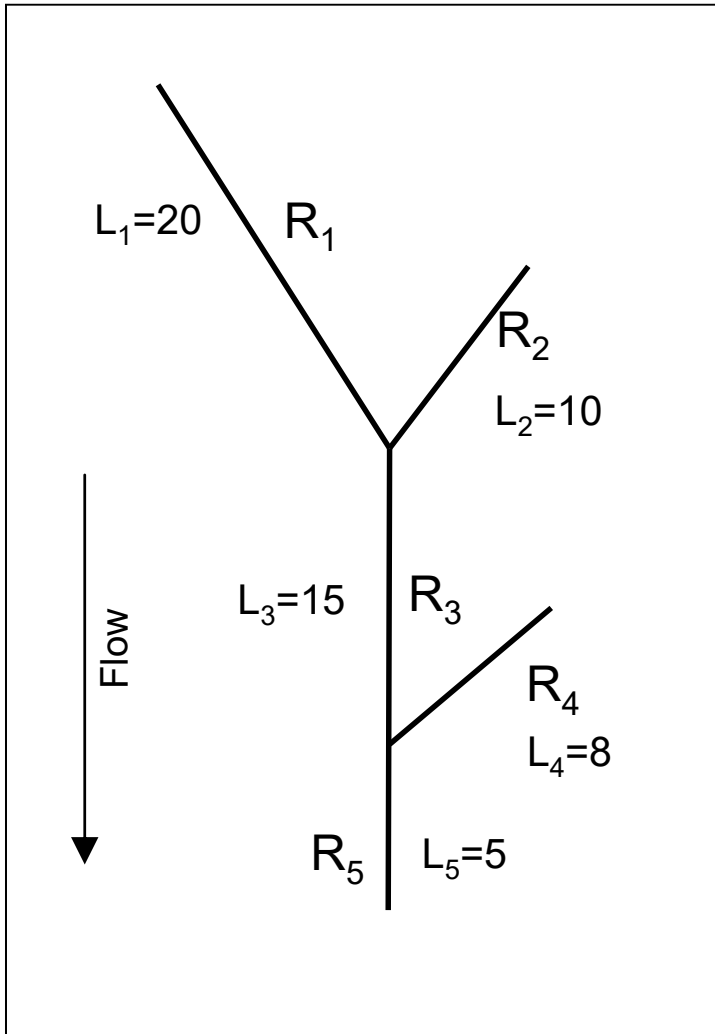
```
graph TD; A[Build the Landscape Network] --> B[Calculate Spatial Data]; B --> C[Assemble the SSN];
```

Build the Landscape Network

Calculate Spatial Data

Assemble the SSN

Calculating Hydrologic Distances



Upstream Distance: Edges

Distance from network outlet to upstream node of each edge

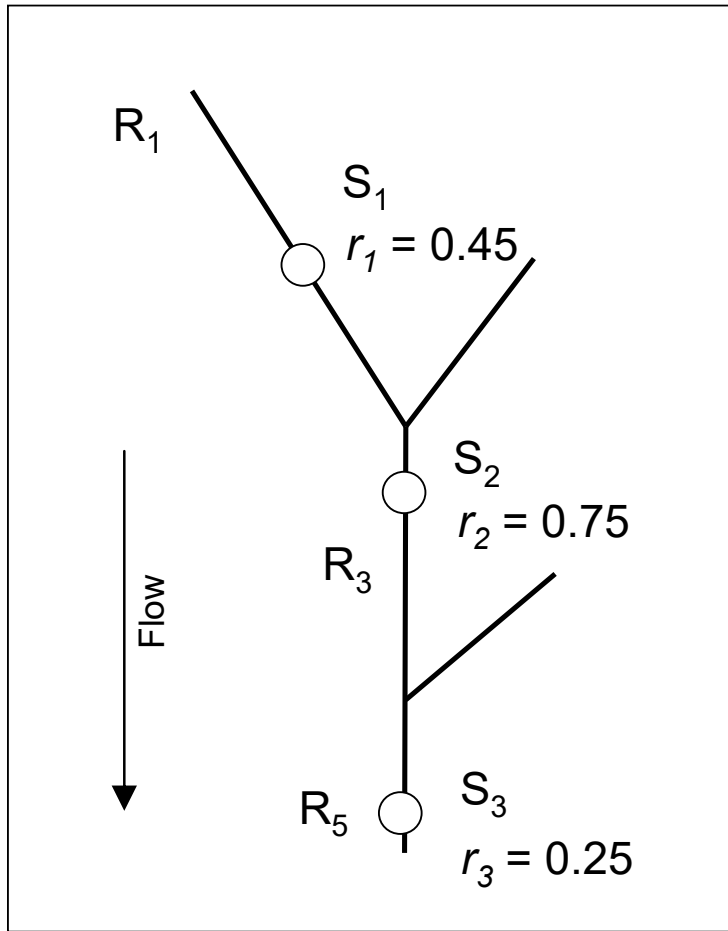
rid (R _j)	Length (L _j)	upDist _j
R ₁	20	40
R ₂	10	30
R ₃	15	20
R ₄	8	13
R ₅	5	5

`updist_edges()`

Calculating Hydrologic Distances

Upstream Distance: Sites

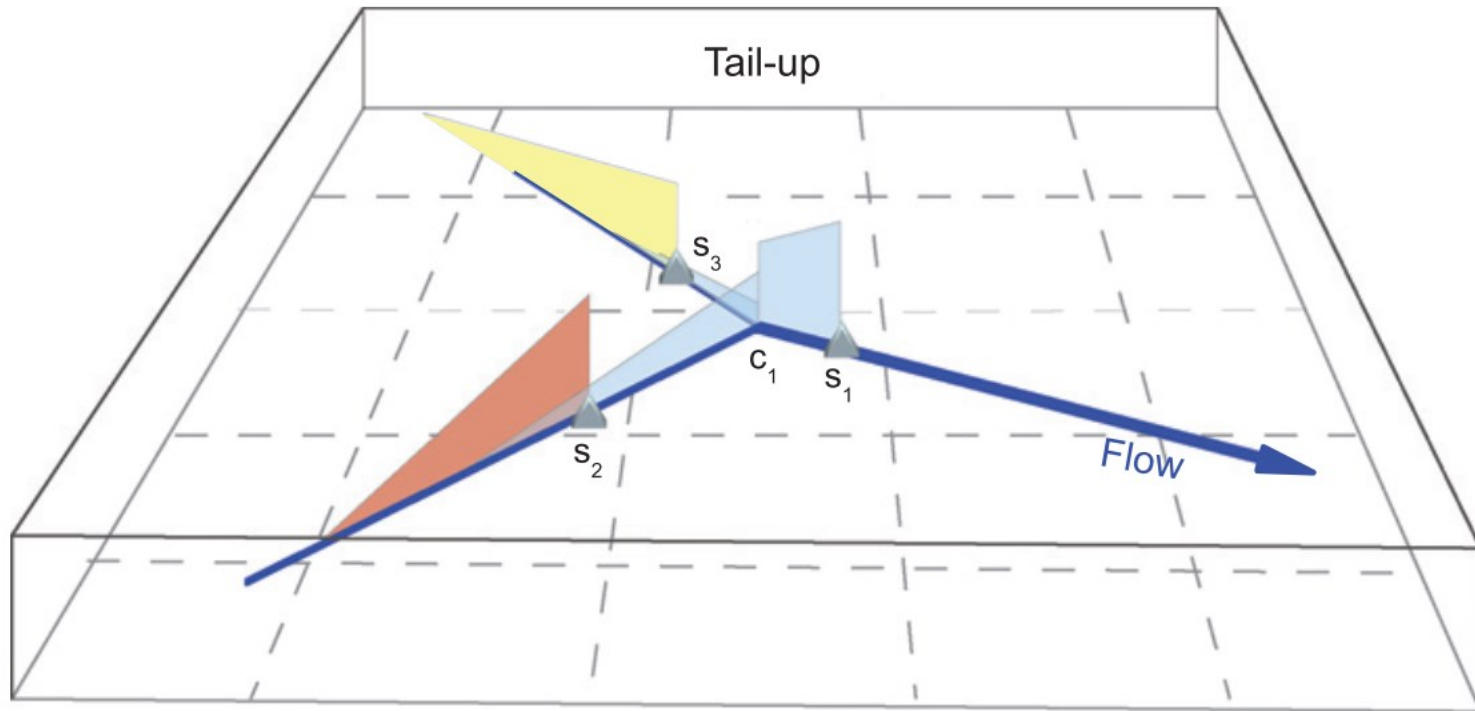
Distance from network outlet to each site



Site ID (S_i)	ratio (r_i)	rid (R_j)	Length (L_j)	upDist (S_i)
S_1	0.45	R_1	20	29.00
S_2	0.75	R_3	15	16.25
S_3	0.25	R_5	5	1.25

`updist_sites()`

Calculating Spatial Weights



Tail-up moving-average function must be split at confluences using spatial weights

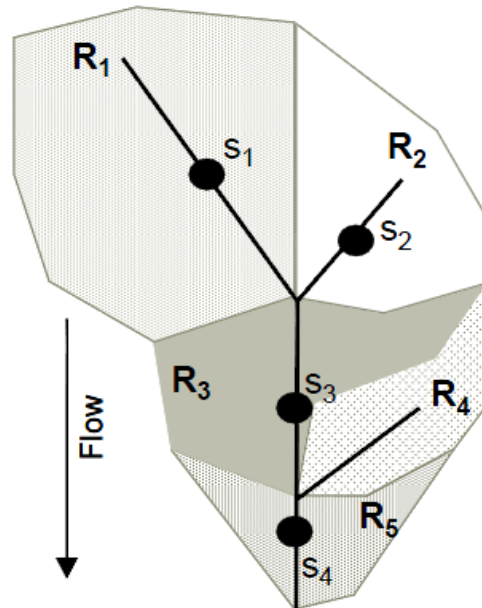
Calculating Spatial Weights

3-Step Process

Segment ID	Watershed Area (q_i)	Segment PI (ω_k)	Segment AFV
R_1	50	0.59	0.5
R_2	35	0.41	0.35
R_3	115	0.85	0.85
R_4	20	0.15	0.15
R_5	160	1	1

$$\omega(R_3) = \frac{q(R_3)}{q(R_3) + q(R_4)} = \frac{115}{115 + 20} = 0.85$$

Segment AFV $R_1 = \omega(R_5) * \omega(R_3) * \omega(R_1) = 1 * 0.85 * 0.59 = 0.50$ (A)



Site ID	Site AFV (Ω_i)
s_1	0.5
s_2	0.35
s_3	0.85
s_4	1

$$\Omega(s_4) = \text{Segment AFV } R_5 = 1$$

(B)

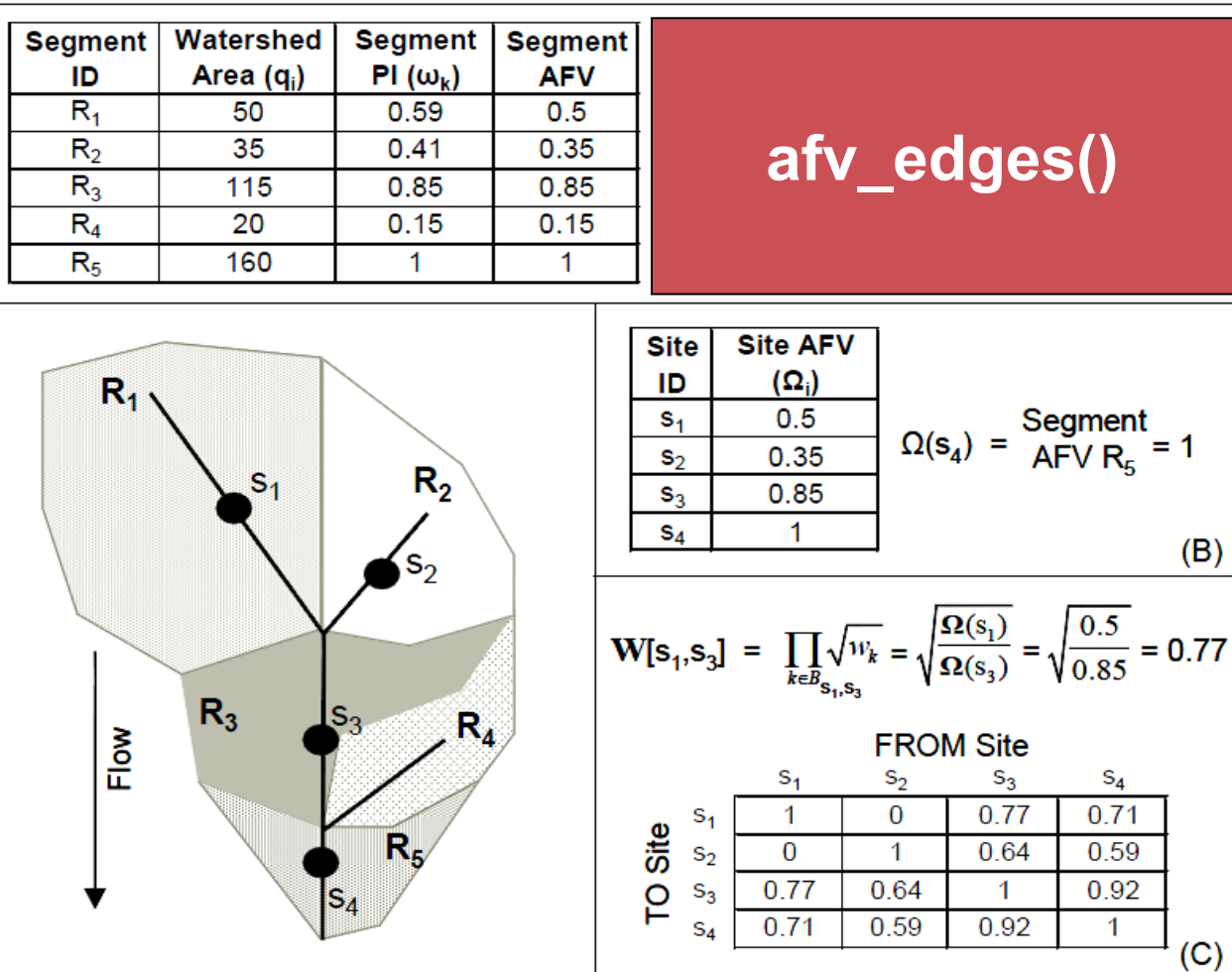
$$W[s_1, s_3] = \prod_{k \in B_{s_1, s_3}} \sqrt{w_k} = \sqrt{\frac{\Omega(s_1)}{\Omega(s_3)}} = \sqrt{\frac{0.5}{0.85}} = 0.77$$

		FROM Site			
		s_1	s_2	s_3	s_4
TO Site	s_1	1	0	0.77	0.71
	s_2	0	1	0.64	0.59
	s_3	0.77	0.64	1	0.92
	s_4	0.71	0.59	0.92	1

(C)

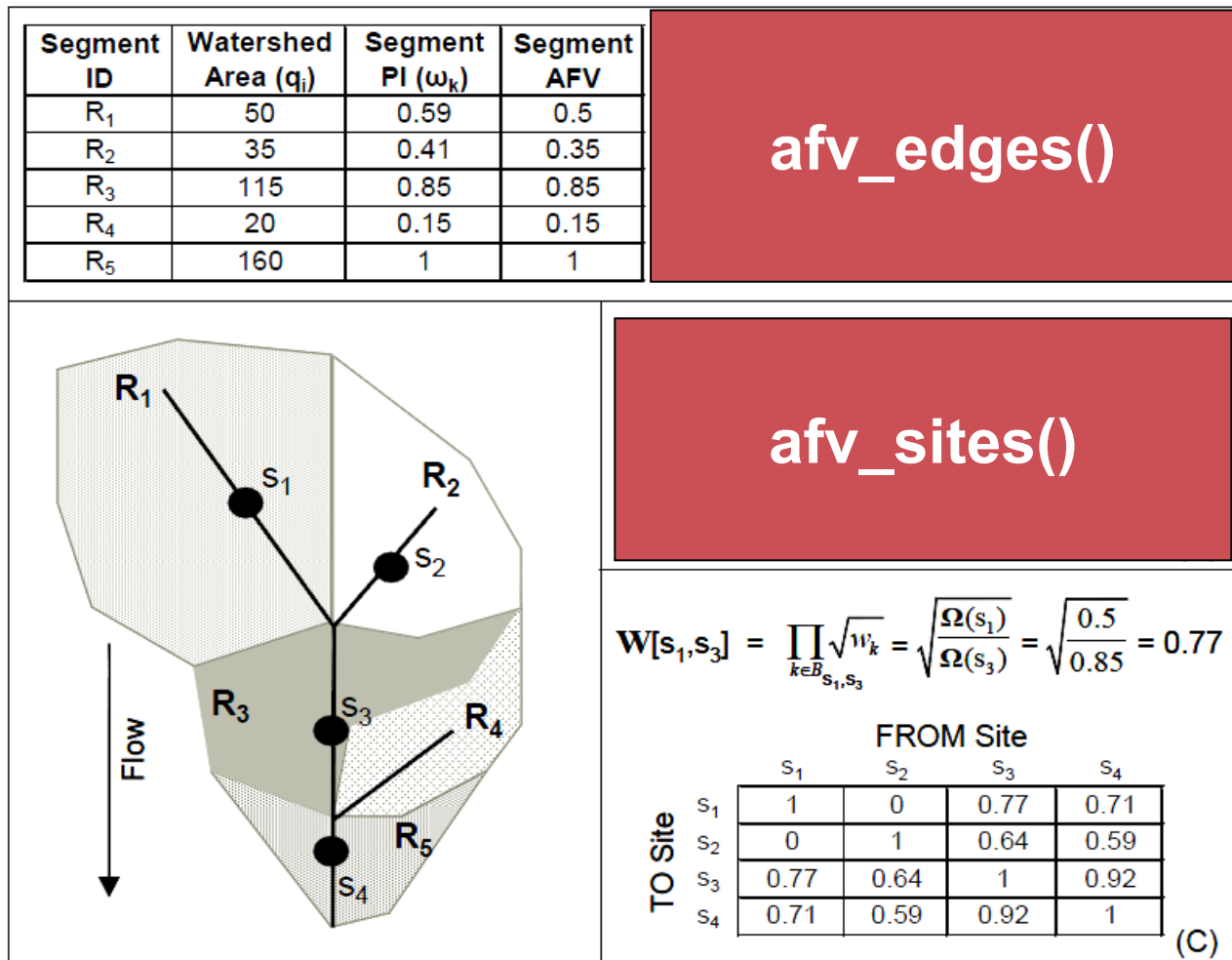
Calculating Spatial Weights

3-Step Process



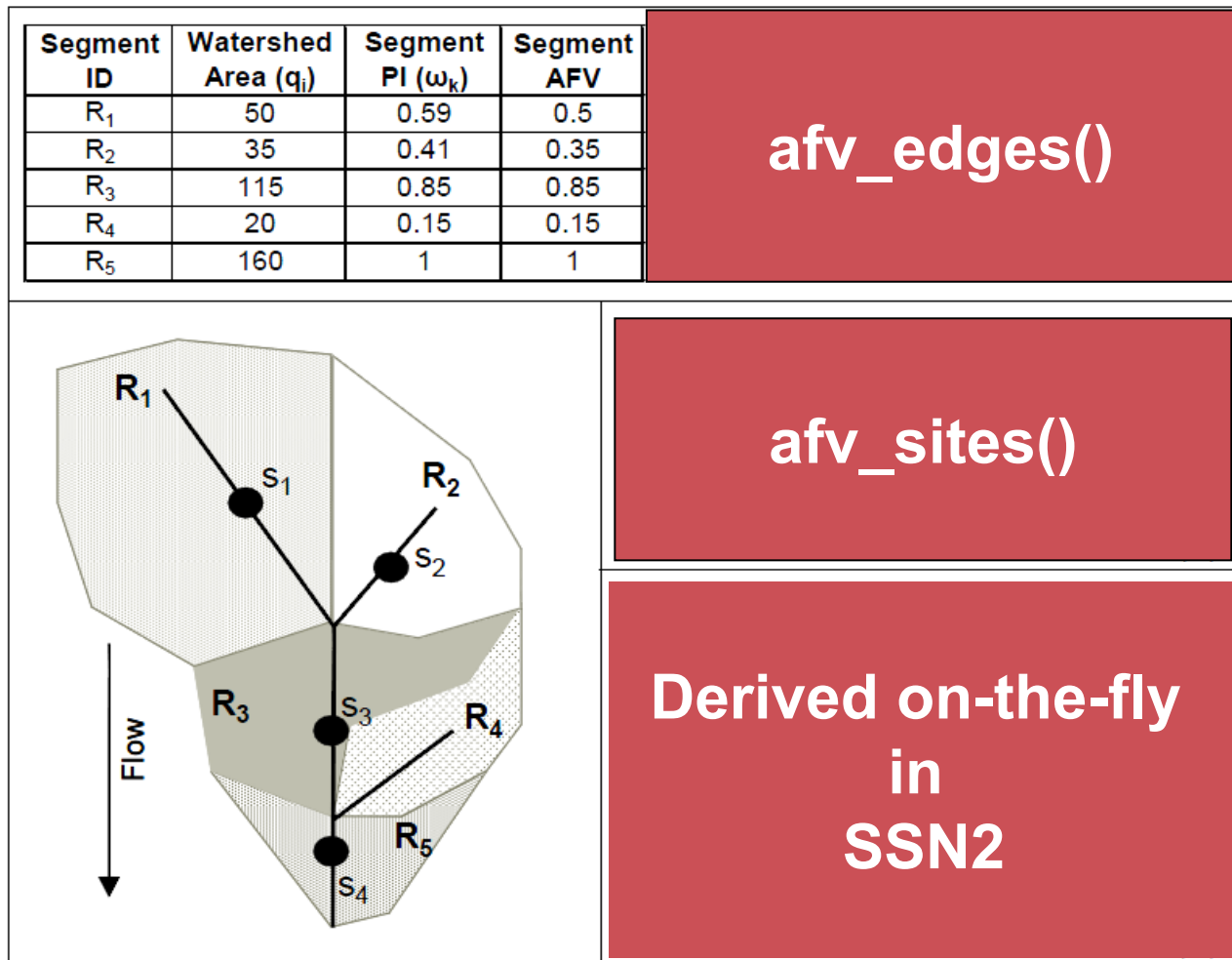
Calculating Spatial Weights

3-Step Process



Calculating Spatial Weights

3-Step Process



Calculating Spatial Weights

Macroinvertebrate Index Model Results

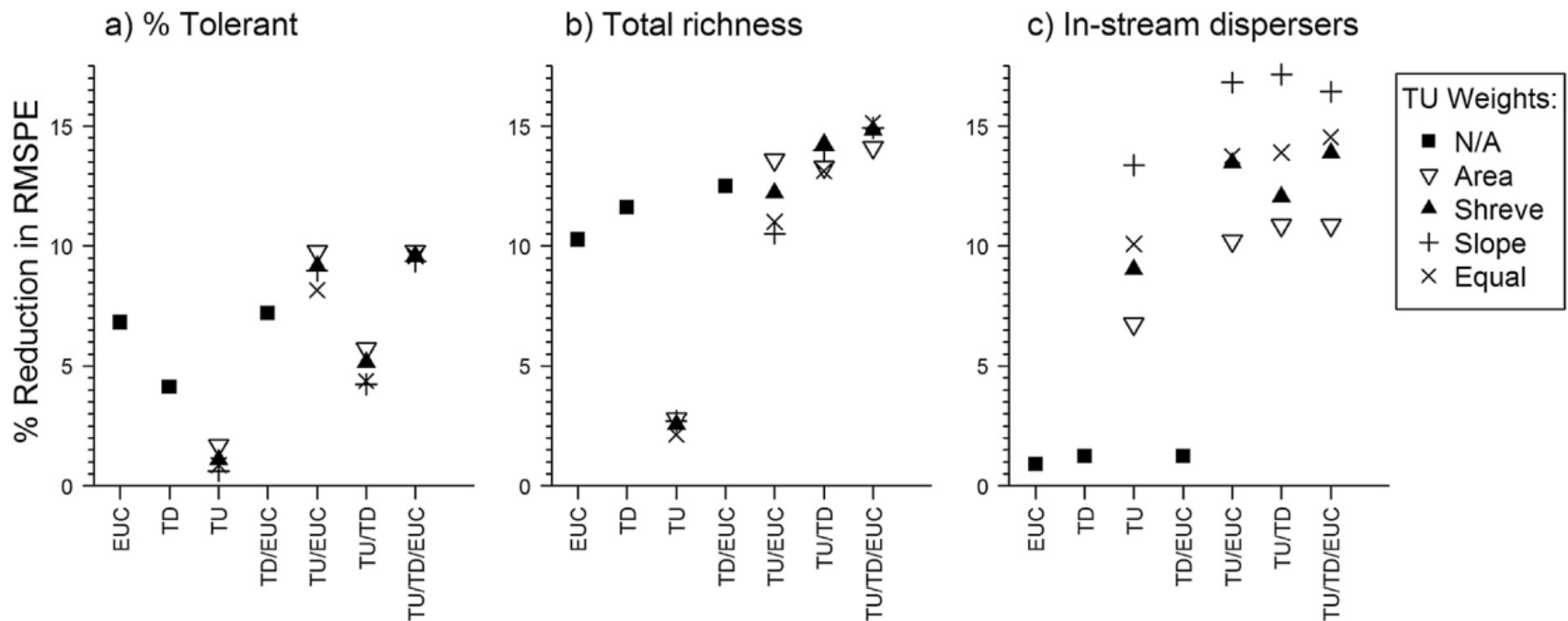


Fig.5: Frieden et al. (2014) Environmental Modelling & Software

Where to find variables for spatial weights

National-scale, “attributed” streams datasets

- US NHDPlus V2 1:100,000 medium resolution data
- US NHDPlus High Resolution 1:24,000 data
- Australian Geofabric Surface Network dataset

Use a simpler weighting scheme

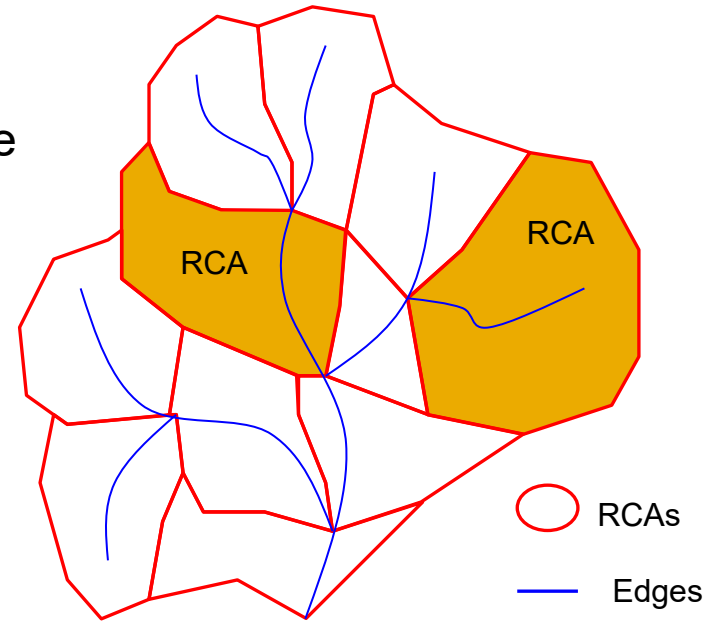
- Equal weights
- Shreve’s stream order (not Strahler) calculated in GIS

Delineate reach contributing areas (RCAs)

- Calculate watershed area, slope, etc.

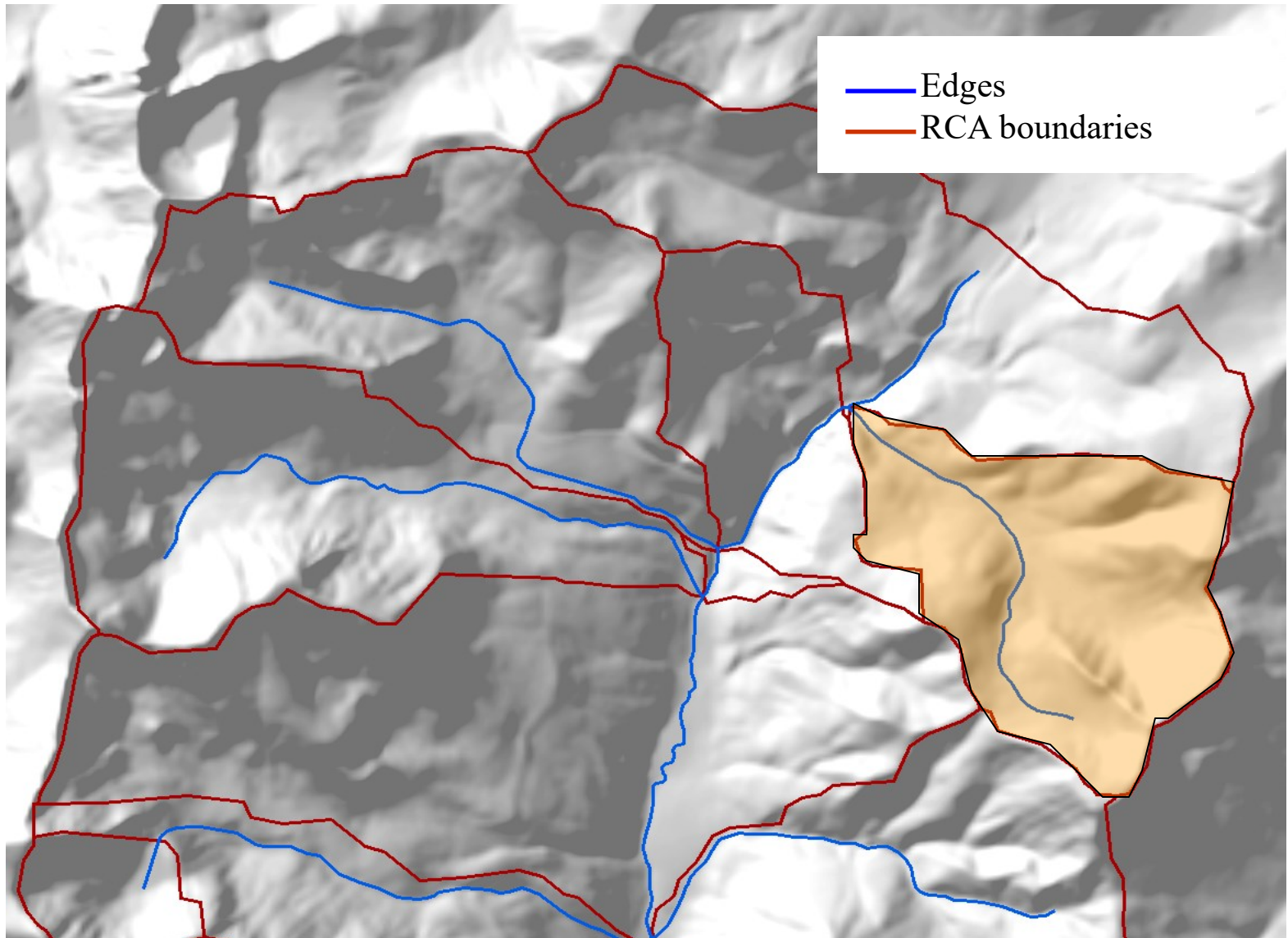
Reach Contributing Areas

- Non-overlapping, contiguous tessellation of RCAs
 - Delineated based on topography
 - Represent land area that *could* contribute overland flow to an edge
- 1 to 1 relationship
 - Edges linked to the landscape
- Summarise RCA attributes
 - Area
 - Topography
 - Land use, soils, geology, vegetation, etc.
- Accumulate RCA attributes downstream
 - Watershed attributes for downstream node of each edge feature

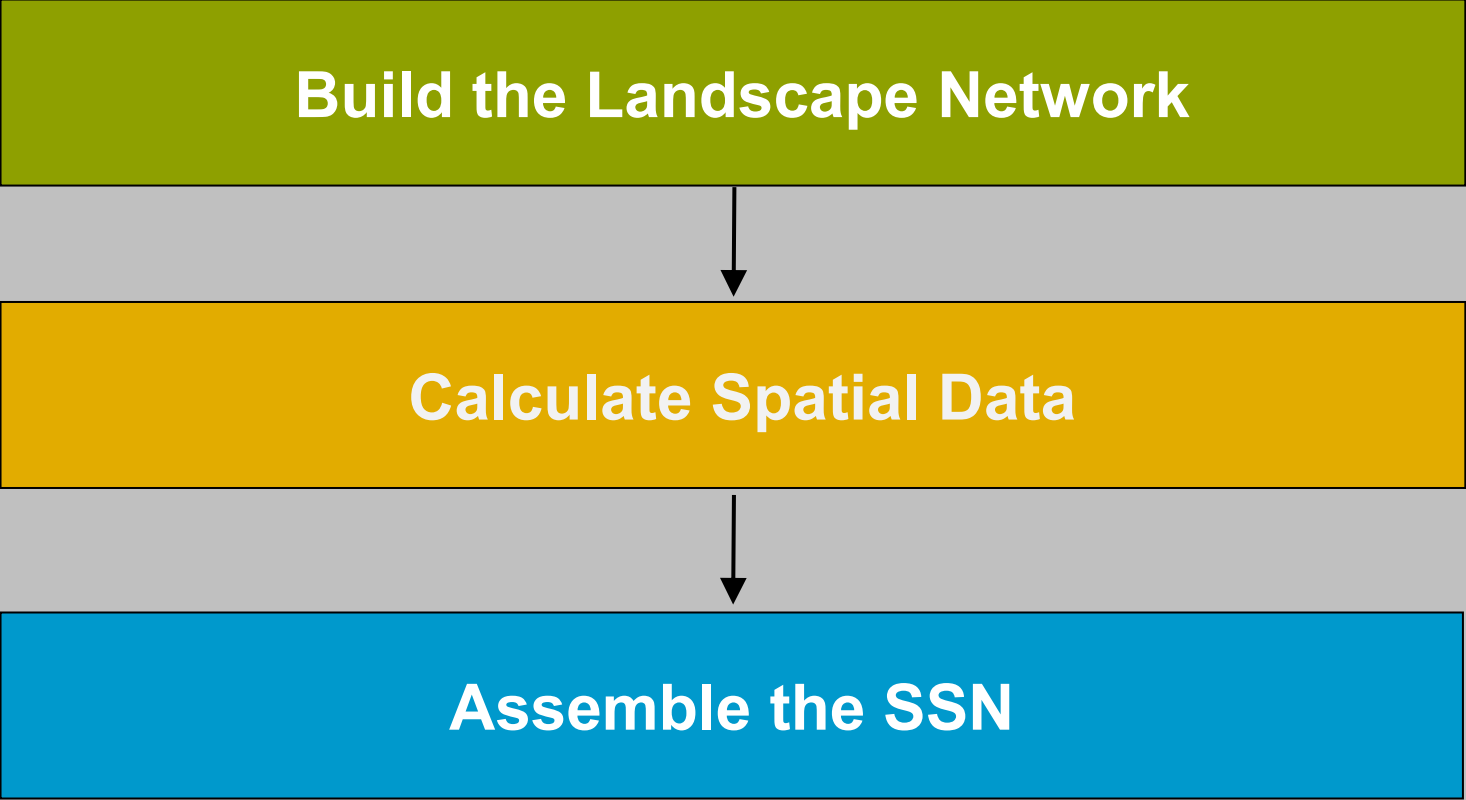


See 'Delineating Reach Contributing Areas Using QGIS' tutorial.

Reach Contributing Areas



Spatial Data Processing Workflow



```
graph TD; A[Build the Landscape Network] --> B[Calculate Spatial Data]; B --> C[Assemble the SSN];
```

Build the Landscape Network

Calculate Spatial Data

Assemble the SSN

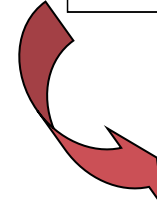
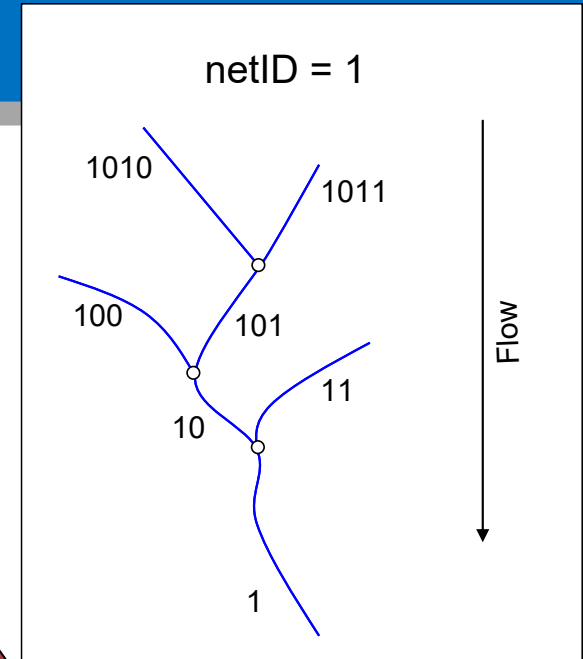
Assemble the SSN

Binary Identifiers

- More efficient for calculating pair-wise distances & spatial weights in R
- Based on unique within-network binary IDs
- netIDx.dat: text file stores topological information for each network

Additional attributes assigned

- Network ID (netID) assigned to edges, sites, and preds
- Point ID (pid) assigned to sites & preds
 - Unique to measurement
- Location ID (locID) assigned to sites & preds
 - Unique to location



netID1.dat

```
"rid", "binaryID"  
418, 1  
409, 10  
410, 11  
407, 100  
408, 101  
176, 110  
405, 111  
347, 1010
```

Assemble the SSN

.ssn folder: Contains topological, spatial, and attribute information needed to fit a spatial statistical model in SSN2

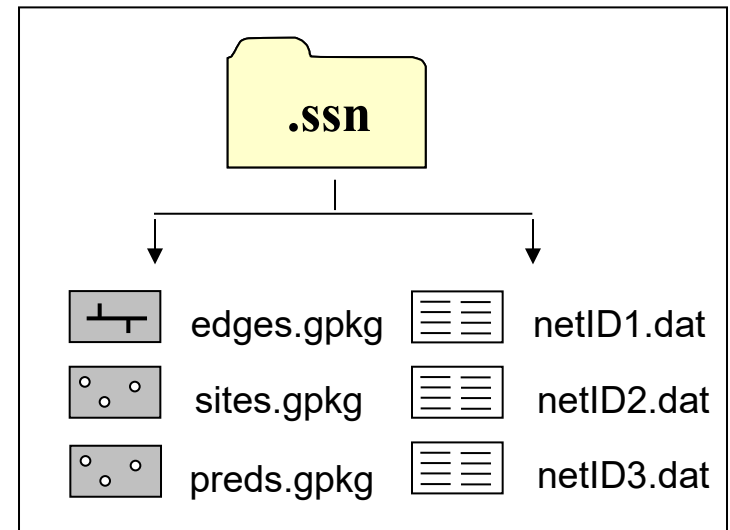
- Reformat information so that it can be easily accessed from R

.ssn directory components:

- Sites & edges GeoPackages
- Text file containing binary IDs for each network in LSN
- Prediction sites GeoPackage: optional

Easily transferable

- If it's in the .ssn folder, you need it



The SSN Object

SSN: Named list of simple features (sf) data.frames

1. **edges:** An sf data.frame containing the stream network
2. **obs:** An sf data.frame containing the observed site locations.
3. **preds:** A named list of sf data.frames containing prediction site locations.
4. **path:** the local filepath for the .ssn directory associated with the SSN object

Check the SSN

- **Local files exist:** .ssn directory and binaryID.db
- **Classes, object structure, geometry types**
- **Topology: netgeom column**

“ENETWORK (netID, rid, upDist)”

“SNETWORK (netID, rid, upDist, ratio, pid, locID)”

- Data present
- NA values
- Duplicates in pid and rid
- Valid data ranges

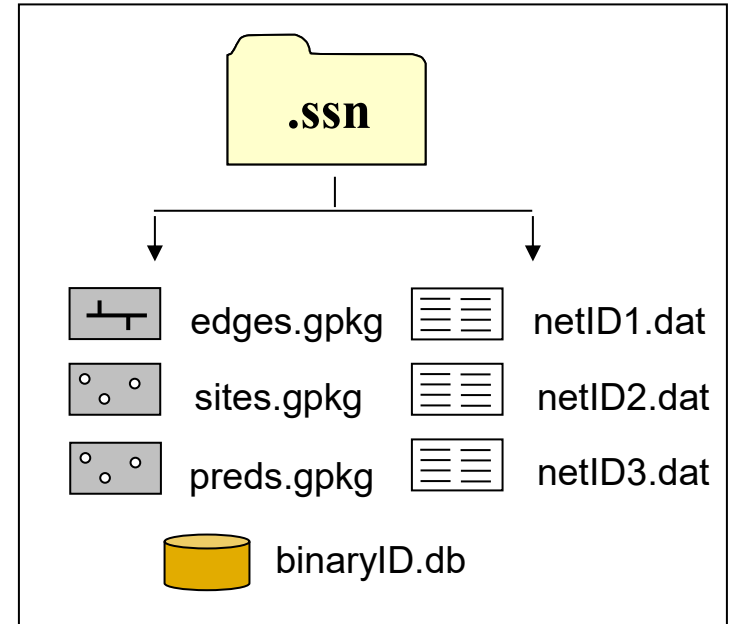
ssn_check()

- **Additive function value columns**
 - Columns present
 - NA values
 - Valid data ranges

Importing a SSN object in SSN2

SSN2 ssn_import

- creates an SSN object
- binaryID.db: SQLite database containing netIDx tables
- Writes to .ssn directory
- ssn\$path: filepath to .ssn directory



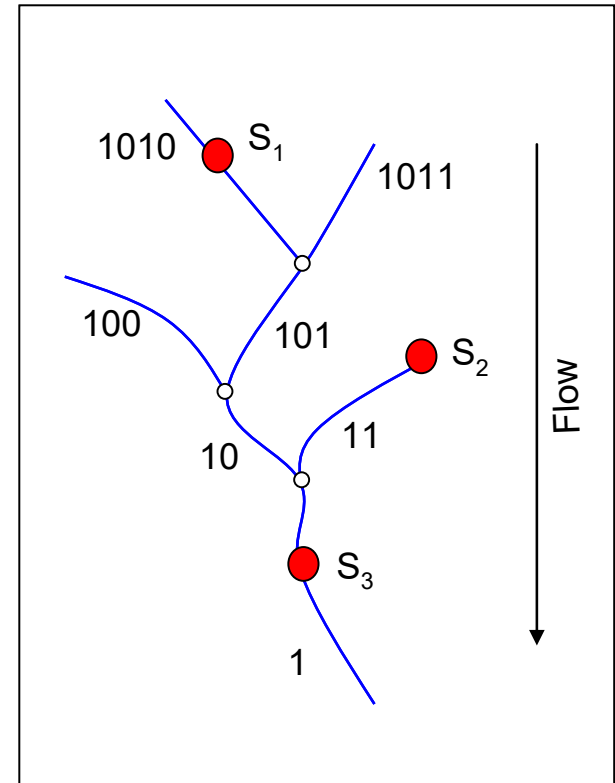
netgeom column

- Character string describing the topology identifiers/columns for each edge or point feature
- Format differs depending on feature geometry:
 - edges: “ENETWORK (netID rid upDist)”
 - point: “SNETWORK (netID rid upDist ratio pid locID)”

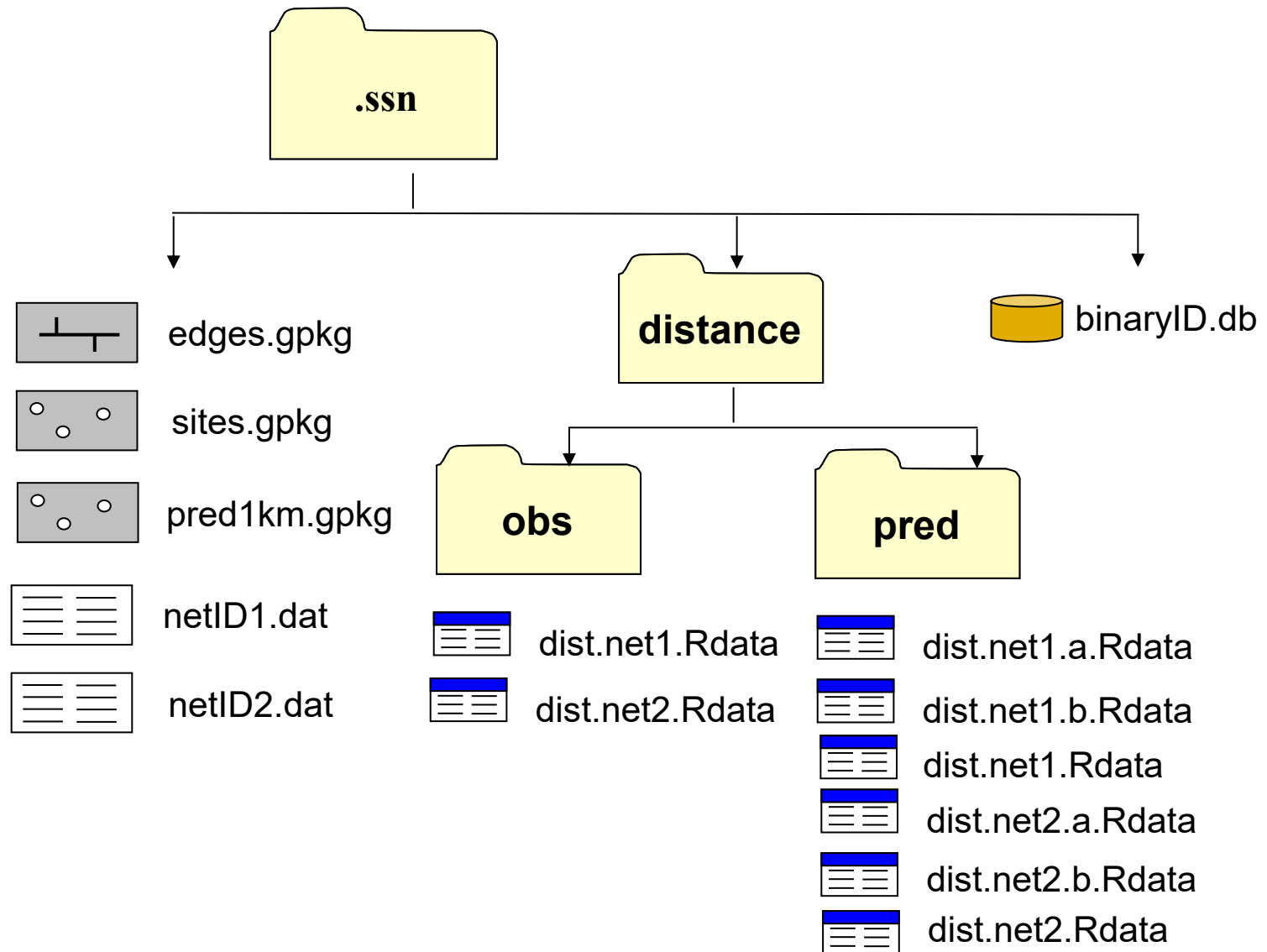
Calculating Hydrologic Distances

ssn_create_distmat()

- Compare netID
- Compare binaryIDs
 - Flow-connected: nested binary IDs
 - Flow-unconnected: binary IDs diverge
- Calculate two hydrologic distance measures using upstream distances for edges & sites
 - Downstream-only hydrologic distance
 - Total hydrologic distance
- More efficient than traversing the network and measuring distances



Calculating Hydrologic Distances



Additional Resources

Lists

Working with lists in R (Equitable Equations)

<https://www.youtube.com/watch?app=desktop&v=X8INTDeiKiE>

sf objects

Spatial Data Science: With Applications in R (Pebesma and Bivand 2023)

<https://r-spatial.org/book/>

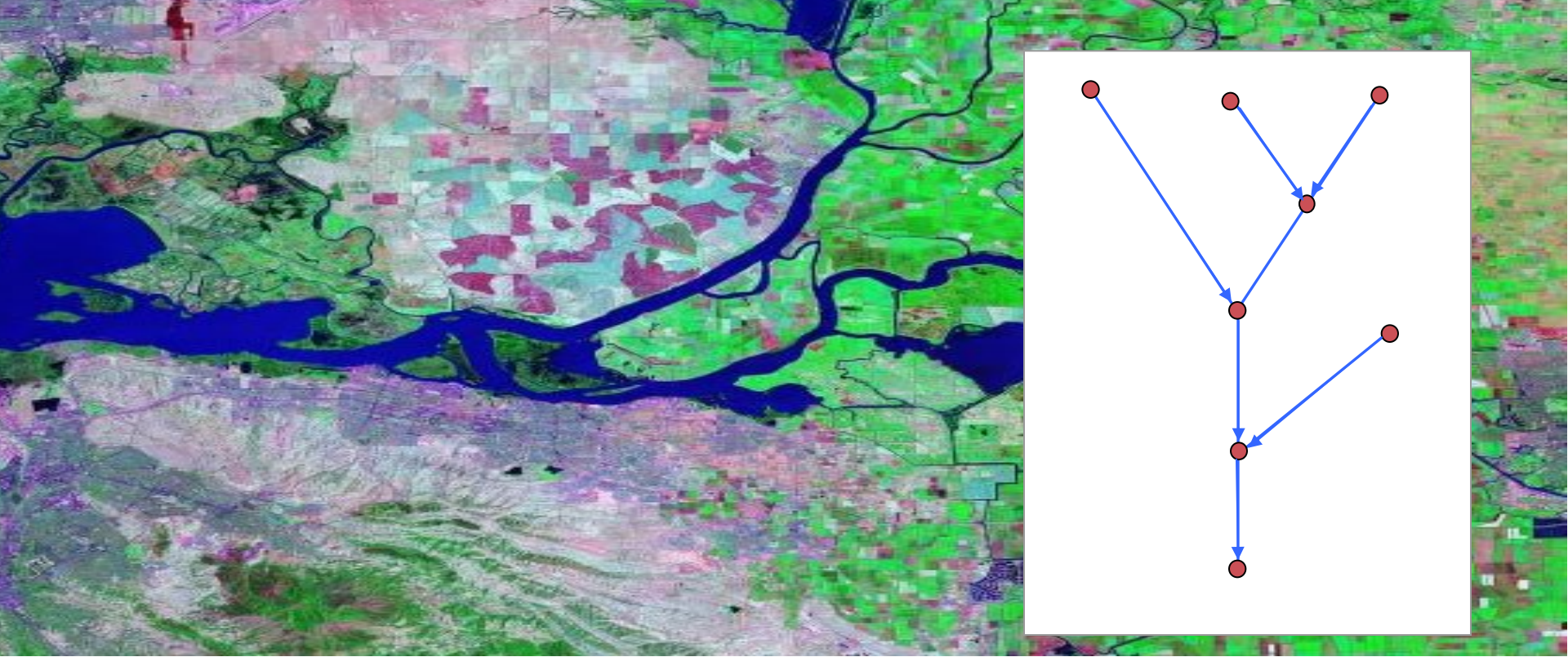
- Chapter 7: sf objects <https://r-spatial.org/book/07-Introsf.html>
- Chapter 8: plotting spatial data <https://r-spatial.org/book/08-Plotting.html>

SSNbler Website and vignettes: <https://pet221.github.io/SSNbler/>

SSN2 Website and vignettes: <https://usepa.github.io/SSN2/>

USFS RMRS: Tutorials, vignettes, example datasets, bibliography

<https://research.fs.usda.gov/rmrs/projects/spatial-stream-network>



Thank-you

Erin E. Peterson

EP Consulting, Brisbane Australia